

Phase 3: Database Creation, Table Population & Business Questions

1. How The Dataset Was Converted Into Tables

Start from raw survey file (CSV)

- We began with the original coffee_survey CSV file containing ~4042 responses and many columns: demographics, coffee habits, preferences, purchase behavior, and ratings for samples A–D.
- Because the file had text, numbers, “NA” strings, and special characters, we first treated it as a raw data source (no normalization yet).

Create a staging table in MySQL (coffee_raw)

- We created a database/schema (coffee_db) and then imported the CSV into a single wide table called coffee_raw.
- All columns were initially stored as TEXT to avoid import errors from non-numeric values like “Less than 1” or “NA”.
- This staging table is an exact copy of the survey file and is only used as the source for our normalized tables.

Analyze variables and design the ERD (logical grouping)

- We reviewed the survey questions and grouped columns into logical themes:
 - **CoffeeHabits** – cups per day, where they drink, brew method, additives, dairy, sweetener, favorite drink
 - **CoffeePreference** – style, strength, roast level, caffeine preference, expertise, reasons for drinking, taste, knowledge of source
 - **PurchaseBehavior** – where they buy coffee, how much they pay or are willing to pay, cafe value rating, equipment spend/value, monthly spend
 - **CoffeeCategoryFeedback** – bitterness, acidity, preference ratings for samples
 - **OverallCoffeeChoice** – overall preferred sample
 - **Participant** – demographic and ID information, plus foreign keys to the other tables

Create normalized tables with primary and foreign keys

- In MySQL six tables were created from the ERD
- Each table has an auto-increment primary key (HabitID, PreferenceID, PurchaseID, FeedbackID, ChoiceID) except Participant, which uses the survey submission_id as ParticipantID.
- Participant contains the demographic attributes and foreign keys to each of the other tables, establishing the one-to-many relationships (one shared habit/preference row can be linked to many participants).

Populate dimension tables from coffee_raw

- While inserting data from raw table into respective tables, we handled data cleaning:
 - Converted “NA” and blank strings to NULL.
 - Converted categorical values (e.g., “Less than 1”, “2–3 cups”) into integers.

Link participants to habits/preferences/behavior/feedback

- Finally, we populated the Participant table by joining coffee_raw to each of the normalized tables on their shared attributes.
- This step links each survey respondent to their corresponding habit, preference, purchase behavior, feedback, and overall choice records.

Result: normalized, query-friendly schema

- The final schema we designed eliminates unnecessary duplication, ensures a consistent structure through relationships and constraints, and helps organize the data so it can be easily queried for business purposes.

2. Challenges Faced During Importing the Data and How we fixed them

Challenge	Description	How it resolved
Large CSV not importing fully (only ~170 rows imported instead of 4042)	MySQL Workbench stopped reading the file due to internal line breaks, commas inside quotes, and mixed data types.	Cleaned the CSV, removed embedded newline characters, converted everything to TEXT, and re-imported using the staging table.
MySQL Workbench Import Wizard errors	The wizard failed or truncated data because of special characters, long text, and 'NA' values.	Bypassed the wizard and created a raw staging table (coffee_raw) using LOAD DATA command to safely import the dataset.
LOAD DATA LOCAL INFILE blocked due to secure_file_priv restriction	MySQL only allowed loading files from a specific secure directory.	Located secure_file_priv using SHOW VARIABLES and placed the cleaned CSV in that folder.
Handling 'NA', blanks, inconsistent values	'NA' and empty strings caused inconsistencies in numeric columns.	Converted these to NULL using COALESCE() function and CASE expressions during insert.
Normalization insert returned 0 rows	Insert statements failed because normalized column names didn't match raw CSV column names.	Correctly mapped each normalized column to coffee_raw and rewrote INSERT statements.
Inconsistent Numerics	Integer fields were imported as Floats or Text, causing errors.	Data was loaded as VARCHAR (text) first, then explicitly cast (e.g., CAST(... AS UNSIGNED)) during insertion into target tables.

Inconsistent Text Values	The cups column had non-standard text (e.g., "1-2 cups", "Three", "none").	SQL CASE statements were used to map these inconsistent text strings to standard, usable integer values (e.g. 1, 3, 0).
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3. Data Dictionary

Sr.No	Variable Name	Data Type	Description
1	ParticipantID	char	Submission ID (represents a unique participant)
2	Age	int	What is your age?
3	CupsPerDay	int	How many cups of coffee do you typically drink per day? (less than 1 represented as 0 and more than 3 as 4)
4	ConsumptionPlace	char	Where do you typically drink coffee?
5	BrewMethod	char	How do you brew coffee at home?
6	PurchaseLocation	char	On the go, where do you typically purchase coffee?
7	FavoriteDrink	char	What is your favorite coffee drink?
8	CoffeeAdditives	char	Do you usually add anything to your coffee?
9	DairyType	char	What kind of dairy do you add?
10	SweetenerType	char	What kind of sugar or sweetener do you add?
11	PreferredStyle	char	Before today's tasting, which of the following best describes what kind of coffee you like?
12	StrengthPreference	char	How strong do you like your coffee?
13	RoastLevel	char	What roast level of coffee do you prefer?
14	Caffeine	char	How much caffeine do you like in your coffee?
15	PriorExpertiseLevel	int	Lastly, how would you rate your own coffee expertise?
16	CoffeeA_Bitterness	int	Coffee A - Bitterness (on a scale of 1-5)
17	CoffeeA_Acidity	int	Coffee A - Acidity (on a scale of 1-5)
18	CoffeeA_Preference	int	Coffee A - Personal Preference (on a scale of 1-5)
19	CoffeeB_Bitterness	int	Coffee B - Bitterness (on a scale of 1-5)
20	CoffeeB_Acidity	int	Coffee B - Acidity (on a scale of 1-5)
21	CoffeeB_Preference	int	Coffee B - Personal Preference (on a scale of 1-5)
22	CoffeeC_Bitterness	int	Coffee C - Bitterness (on a scale of 1-5)
23	CoffeeC_Acidity	int	Coffee C - Acidity (on a scale of 1-5)
24	CoffeeC_Preference	int	Coffee C - Personal Preference (on a scale of 1-5)
25	CoffeeD_Bitterness	int	Coffee D - Bitterness (on a scale of 1-5)
26	CoffeeD_Acidity	int	Coffee D - Acidity (on a scale of 1-5)
27	CoffeeD_Preference	int	Coffee D - Personal Preference (on a scale of 1-5)
28	OverallPreferredSample	char	Lastly, what was your favorite overall coffee? (from A, B, C, D)
29	WorkFromHome	char	Do you work from home or in person?
30	SpentPerMonth	int	In total, much money do you typically spend on coffee in a month?

31	ReasonForDrinking	char	Why do you drink coffee?
32	LikeTaste	char	Do you like the taste of coffee?
33	CoffeeSource	char	Do you know where your coffee comes from?
34	MaxPaidPerCup	int	What is the most you've ever paid for a cup of coffee?
35	WillingToPay	char	What is the most you'd ever be willing to pay for a cup of coffee?
36	CaféValueRating	char	Do you feel like you're getting good value for your money when you buy coffee at a cafe?
37	EquipmentSpend	int	Approximately how much have you spent on coffee equipment in the past 5 years?
38	EquipmentValueRating	char	Do you feel like you're getting good value for your money from your coffee equipment?
39	Gender	char	What is your gender?
40	EducationLevel	char	What is your educational qualification?
41	Ethnicity	char	What is your ethnic background?
42	EmploymentStatus	char	What is your current employment status?
43	NumOfChildren	int	Do you have children? If yes, how many? (more than 3 is represented as 4)
44	PoliticalAffiliation	char	How would you describe your political preference or affiliation?
45	HabitID	int	Auto-incrementing ID that uniquely represents a habit
46	PreferenceID	int	Auto-incrementing ID that uniquely represents a preference
47	PurchaseID	int	Auto-incrementing ID that uniquely represents a purchase
48	ChoiceID	int	Auto-incrementing ID that uniquely represents a choice
49	FeedbackID	int	Auto-incrementing ID that uniquely represents a feedback

4. Business Questions

1. How many people participated in the coffee survey?
2. What is the average number of cups of coffee consumed per day across participants?
3. Which coffee sample (A, B, C, or D) received the highest average preference score?
4. What is the relationship between the reason for drinking coffee (energy, taste, habit) and willingness to pay more for a cup?
5. Does working from home influence how many cups of coffee a person drinks per day?
6. Which purchase location leads to the highest monthly spending on coffee?
7. How does age group influence willingness to pay for a premium cup of coffee?
8. How do purchase behaviors (monthly spend, max paid per cup, equipment spending, purchase location) vary across different brew-method users?
9. Do health-motivated drinkers (reason for drinking) show different consumption patterns (cups/day, additives used) compared to taste- or energy-motivated drinkers?
10. Which type of drinker (based on cups/day) is most sensitive to price?
11. How does the number of children of a participant affect their overall coffee behavior — including daily coffee consumption, monthly spending on coffee, and their final preferred coffee sample (A, B, C, or D)?